Name: _____

TOPIC TEST 1

Chapter 3: Functions and sketching graphs

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks).
- Part B: 8 short-answer questions (16 marks).
- Part C: 2 analysis questions (6 marks).

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 What is the composite function $f[g(x)]$ given that $f(x) = \ln(x)$ and $g(x) = e^{5x+1}$?

A $5x + 1$

B $e^{5\ln(x)+1}$

C $\ln(5x + 1)$

D $\ln(e^{5x} + 1)$

E $e^{5x} + 1$

[1 mark]

2 Which of the following relations is **NOT** a function?

A $f(x) = x - 2$

B $f(x) = e^{x+1}$

C $f(x) = \dfrac{2}{x + 1}$

D $f(x) = \ln(x - 2)$

E $f(x) = \pm\sqrt{x - 2}$

[1 mark]

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3 Write the inverse function of $y = \frac{1}{2}x - 3$.

A $y = 2x - 3$

B $y = 2x + 3$

C $y = \dfrac{x + 3}{2}$

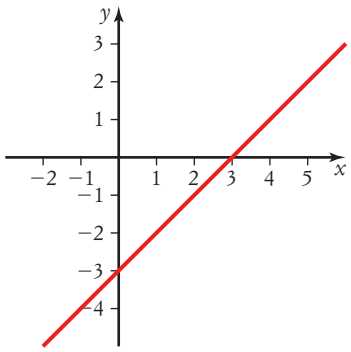
D $y = 2x - 6$

E $y = 2x + 6$

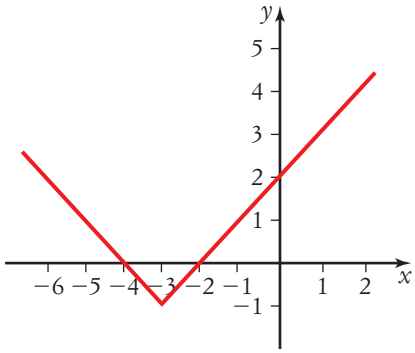
[1 mark]

4 Sketch the function $y = |x - 3| - 1$.

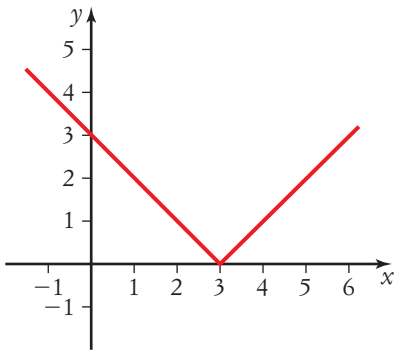
A



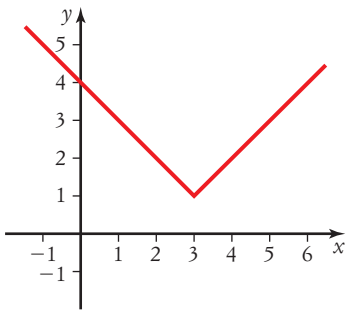
D



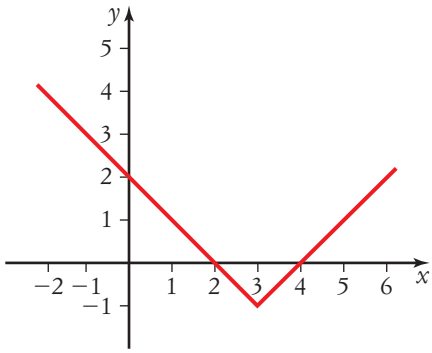
B



E



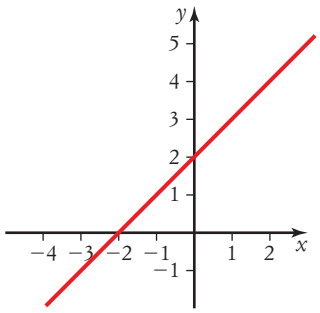
C



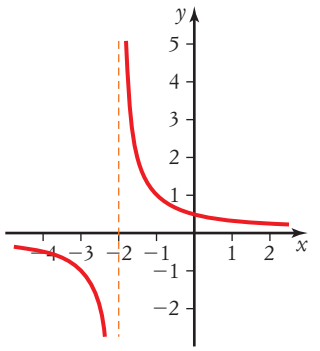
[1 mark]

5 Which graph represents the reciprocal function of $f(x) = (x + 2)$?

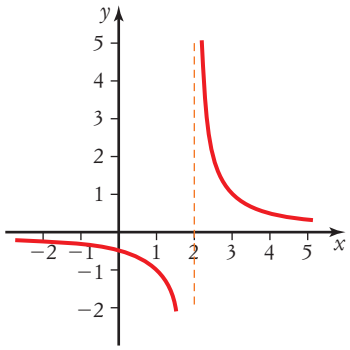
A



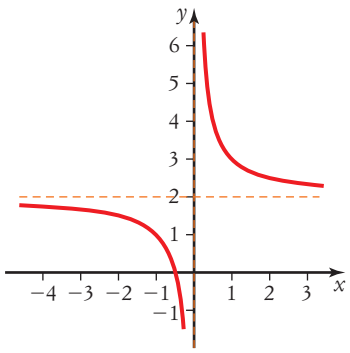
B



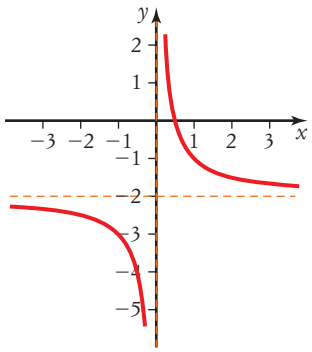
C



D



E



[1 mark]

Part B

Short-answer questions

8 short-answer questions
16 marks
Show your working where appropriate.

1 Sketch the graph of $y = f(|x|)$ given that $f(x) = -x^2 + 2x$.

[2 marks]

2 Sketch the graph of $y = \dfrac{1}{f(x)}$ given that $f(x) = -x^2 + 2x$. Write the domain of $f(x)$.

[2 marks]

3 Sketch the graph of $f(x) = 2|x| - 1$. Write the range of $f(x)$.

[2 marks]

4 Sketch the graph of $f(x) = \dfrac{1}{x-1}$. Write the domain of $f(x)$.

[2 marks]

5 Sketch the graph of $f(x) = \dfrac{x}{x-1}$. Write the range of $f(x)$.

[2 marks]

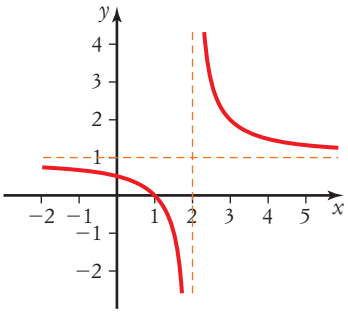
6 Sketch the graph of $f(x) = \dfrac{x^2}{x+1}$. Write the domain of $f(x)$.

[2 marks]

7 Sketch the graph of $f(x) = \dfrac{x^2 + 2x - 1}{x+1}$.

[2 marks]

8 Write a rational function for the following graph.



[2 marks]

Part C

Analysis questions

2 analysis questions
6 marks
Show your working where appropriate.

1 Solve the inequality, $\frac{2x-3}{x-2} > 3$ by sketching functions.

[3 marks]

2 The time (t) it takes for sound to travel one kilometre can be modelled by $t = \frac{1000}{0.6T + 331}$, where T is the air temperature in degrees Celsius and t is in seconds.

a How long does it take for sound to travel 10 kilometres when the air temperature is 25°C?

b Graph the given function to estimate the air temperature when it takes 3 seconds to hear a lightning strike 1 kilometre away.

[3 marks]

[Total marks: 27]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 $f[g(x)] = \ln(e^{5x+1}) = (5x+1)\ln(e) = 5x+1,$

$\therefore$ **A** **[1 mark]**

2 $y^2 = x-2,\ x = y^2+2,\ \therefore f^{-1}(x) = x^2+2,$

$\therefore$ **E** **[1 mark]**

3 $y = \frac{1}{2}x - 3,\ 2(y+3) = x,\ \therefore f^{-1}(x) = 2x+6,$

$\therefore$ **E** **[1 mark]**

4 $y = |x-3| - 1,$

$\therefore$ **C** **[1 mark]**

5 $f(x) = \frac{1}{(x+2)},$

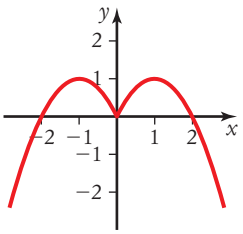
$\therefore$ **B** **[1 mark]**

Part B – Short-answer questions

1 $f(x) = -|x|^2 + 2|x|$

$f(x) = -x(x-2)$

Intercepts at $(-2, 0)$, $(0,0)$ and $(2,0)$ **[1 mark]**

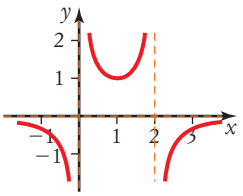


[1 mark]

[2 marks altogether]

2 $y = \frac{1}{f(x)} = \frac{1}{-x^2+2x} = \frac{-1}{x(x-2)}$

Domain: $x \in \mathbf{R},\ x \neq 0,2$ **[1 mark]**

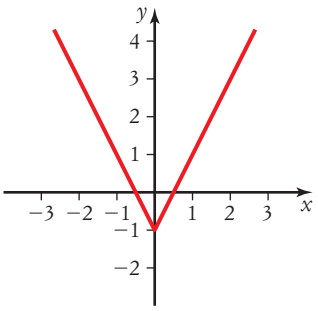


[1 mark]

[2 marks altogether]

3 $f(x) = 2|x| - 1$,

Range: $y > -1$ **[1 mark]**

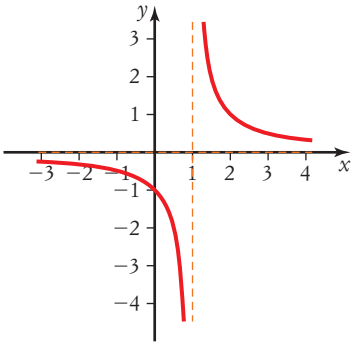


[1 mark]

[2 marks altogether]

4 $f(x) = \dfrac{1}{x - 1}$,

Domain: $x \in \mathbf{R}, x \neq 1$ **[1 mark]**

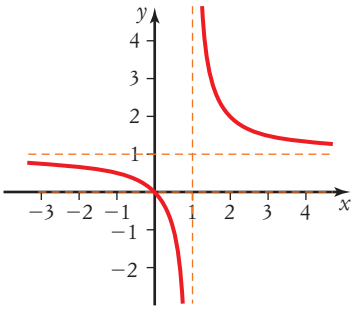


[1 mark]

[2 marks altogether]

5 $f(x) = \dfrac{x}{x - 1}$,

Range: $y \in \mathbf{R}, y \neq 1$ **[1 mark]**

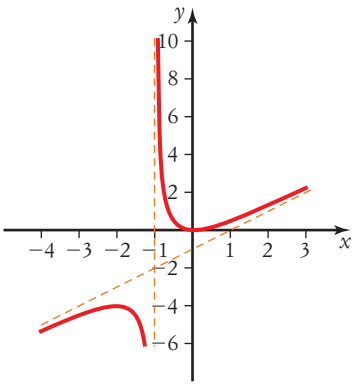


[1 mark]

[2 marks altogether]

6 $f(x) = \dfrac{x^2}{x+1}$,

Domain: $x \in \mathbf{R},\ x \neq -1$ **[1 mark]**

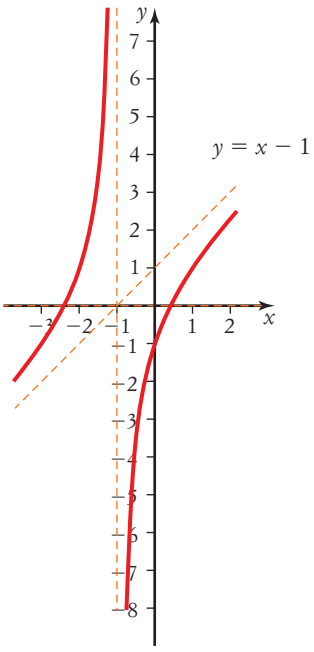


[1 mark]

[2 marks altogether]

7 $f(x) = \dfrac{x^2 + 2x - 1}{x+1}$

Asymptotes **[1 mark]**



[1 mark]

[2 marks altogether]

8 $(y-1)(x-2) = c$,

substitute $(1, 0)$, **[1 mark]**

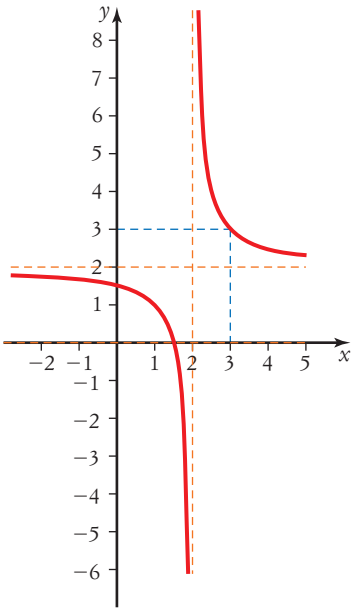
$\therefore c = 1,\ \ (y-1)(x-2) = 1$

or $y = 1 + \dfrac{1}{x-2} = \dfrac{x-1}{x-2}$. **[1 mark]**

[2 marks altogether]

Part C – Analysis questions

1 $\dfrac{2x-3}{x-2} > 3,$ **[1 mark]**



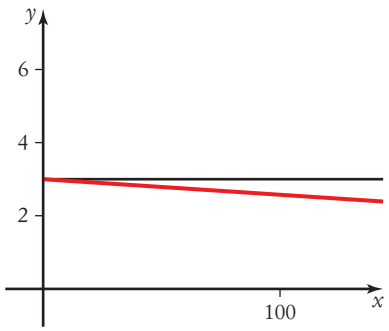
[1 mark]

Solution is $2 < x < 3$. **[1 mark]**

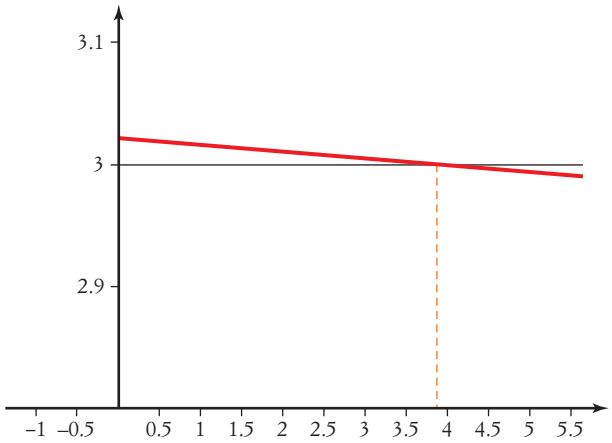
[3 marks altogether]

2 a $t = \dfrac{1000}{0.6(25) + 331} \times 10 \approx 28.9$ seconds **[1 mark]**

b



Looking more closely,



[1 mark]

The air temperature is approximately 3.8°C. **[1 mark]**

[3 marks altogether]